Byte to Bite: Estimating Calories with Vision and Depth

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***Abstract*—Calorie estimation through use of image processing techniques is a complex (but useful) problem to solve. The subject of this report is calorie estimation through the use of single overhead images sourced from a subset of the Nutrition5K dataset. This dataset contains full RGB colour images and depth map data. Several variants of CNN architecture were explored, with a final variant harnessing Cross-Modal Attention and auxiliary depth heads. The model's performance is further enhanced through image pre-processing and other techniques.**

On 5-fold stratified Cross-Validation, the model attains a mean validation set MSE of 4993.62, with the model performing strongly on most dishes (except for high-calorie dishes containing many different foods).

The findings presented in this report demonstrate that with careful selection of a model, combined with proper pre-processing techniques, it is possible to estimate the caloric content of a dish to a degree where it is accurate enough to be useful to those who are monitoring their calorie consumption for health purposes.

***Index Terms*—cross-modal attention, calorie estimation, outlier handling, computer vision**

I. Introduction

In recent decades in industrialised countries, there has been a marked shift in the types of food that are typically consumed [1]. Typical diets are now less healthy [2], which has led to an increase in health problems - including hypertensity and obesity [1]. Consequently, many people have become conscious of their calorie consumption and attempt to estimate their daily caloric intake.

Conventional calorie estimation (e.g. through food logging) is a complex and intensive process - with many people simply being too busy. Using computer vision techniques offers an innovative solution - estimate the calorie content of foods using image data.

This report explores the use of artificial intelligence (AI) models to estimate the calorie content of various dishes solely based on image (and depth map) data. An outline of the methodology is provided, including model design and preprocessing steps. Reasons for those design choices are also included. Results are presented, with some analysis to provide context, followed by a brief review of the key points.

II. Dataset Analysis

The dataset used in this report is a subset of the Nutrition5K dataset. Images were all captured from overhead using an Intel RealSense camera [3]. There are 3,300 training and 189 test instances respectively. For each instance, a raw colour image (see figure 1), a depth map and a colorised depth map were provided. In general, the data appears to contain a variety of meal types (breakfast, dessert, snack etc.) and the selection of food is quite broad. Some instances only contain a single food (e.g. plate of grapes), but the majority contain multiple food type (e.g. plate of beef, with potatoes and carrot).

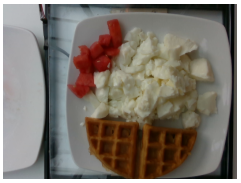

Fig. 1. Example of a typical training instance

Regarding the training data, it was positively skewed; mean = 237.30 Calories, standard deviation = 221.32 Calories, min = 0 Calorie, 25th percentile = 60.84 Calories, median = 186.82 Calories, 75th percentile = 359.28 Calories, max = 3943.33 Calories (see figure 2) [1].

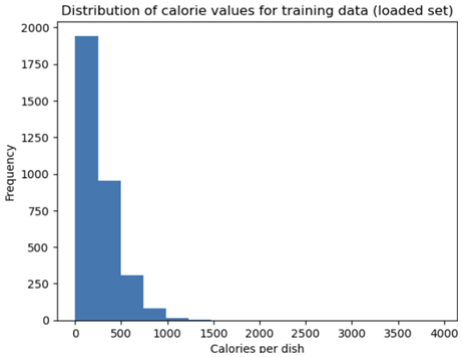

Fig. 2. Distribution of training data

III. Pre-processing

Regarding image resizing, the team settled on a final size of 128×96 pixels, this reduced preprocessing time substantially (from 1 hour down to 2 minutes), while only impacting performance slightly (∼250 increase in MSE for final model).

After testing various combinations of the three data types provided (full colour, depth map, colourised depth), the team found that the optimal combination was the use of the RGB channels combined with the raw depth map (as a 4th channel). Using the colourised depth was a less efficient medium for encoding depth data. Raw depth values were rescaled using min-max scaling, with training and test data being rescaled separately to avoid data leakage. Outliers were removed, these were 18 dishes with very high calorie counts (see error analysis

[1]Calorie (with capital c) is used interchangeably to mean 1,000 calories (i.e. 1 kcal, lowercase c)

section), in addition to a training image (dish_2368) which was removed due to a missing depth map.

Regarding data augmentation, three techniques were evaluated, with the hope that model performance could be made more robust. These were gamma correction and linear brightness/contrast adjustment (for colour channels) and mirroring images (horizontally and/or vertically). It was found that only augmentation by mirroring the images resulted in any improvement, leading to the others being discarded as they actually worsened performance ($\sim$1,500 MSE increase for final model).

IV. Models

To ensure robust results, all instances were shuffled before applying 5-fold stratified Cross Validation. Three models were evaluated, with team iteratively improving the design. Given the limited amount of compute available and use of cross-validation, the team had to carefully select which hyperparameters were tested.

A. Standard Convolutional Neural Network

It was decided to initially begin using a standard convolutional neural network (CNN), which possesses several key advantages over a standard (dense) neural network. Crucially, CNNs are able to extract features automatically (convolutional layers learn features) and possess translation invariance through the use of pooling and convolution operations. This initial model provided an insightful baseline for subsequent work.

It used a classic CNN design with every second convolutional layer being followed by a pooling layer (see table 1), with 303,937 parameters in total (with best hyperparameters). The hyperparameters tested were filter size (3, 5) and Dense layer size (32, 128, 256), with 3 and 128 performing best for each hyperparameter respectively. The number of epochs (60) and the learning rate (1e-3) were kept constant.

TABLE I
STANDARD CNN

Layer	Layer Details
Convolutional Layer	(3×3 kernels), 32 filters (ReLU)
Convolutional Layer	(3×3 kernels), 32 filters (ReLU)
MaxPooling Layer	2×2
Convolutional Layer	(3×3 kernels), 64 filters (ReLU)
Convolutional Layer	(3×3 kernels), 64 filters (ReLU)
MaxPooling Layer	2×2
Convolutional Layer	(3×3 kernels), 128 filters (ReLU)
Convolutional Layer	(3×3 kernels), 128 filters (ReLU)
GlobalAveragePooling Layer	1×1
Dense Layer	128 Neurons (ReLU)
Output Layer	1 Neuron

Despite its simplicity (compared to later models), the plain CNN architecture performed well (MSE 16,057.78). However, it was hampered by the fact that it treated depth as merely being a 4th image channel and its older architecture (e.g. lack of residual connections).

B. Convolutional Neural Network with Residual Connections

Residual connections were invented to help mitigate the *exploding/vanishing* gradient problem, as deep networks have issues with propagating gradients [4]. They are also referred to as skip connections, permitting the activations of an earlier layer to be fed to a later layer. Moreover, the network learns the residual mapping $F(x) = H(x) - x$ (x is skip connection input), which is often smaller and easier to learn than the initial mapping $H(x)$.

Even for a network that is not deep, residual blocks can improve performance for the above reason. The team chose to add residual blocks as it provided a way to *potentially* improve performance without requiring large amounts of extra compute.

For this second model, the hyperparameters tested were initial number of filters (16, 32, 48), 32 performing best and number of residual layers (2, 4, 8), with 4 layers in total (2x2 blocks) performing the best. The number of epochs and learning rate was left unchanged from the previous model.

The model had a size of 885,809 parameters (see table II) and performance greatly improved (MSE 10,654.38) compared to the standard CNN, indicating that residual blocks strengthened the model's predictive power.

TABLE II
CNN WITH RESIDUAL CONNECTIONS

Layer	Layer Details
Convolutional Layer	(3×3 kernels), 32 filters (BatchNorm + ReLU)
Convolutional Layer	(3×3 kernels), 32 filters (BatchNorm + ReLU)
MaxPooling Layer	2×2
Convolutional Layer	(3×3 kernels), 64 filters (BatchNorm + ReLU)
Residual Block ×2	(3×3 kernels), 64 filters (BatchNorm + ReLU)
MaxPooling Layer	2×2
Convolutional Layer	(3×3 kernels), 128 filters (BatchNorm + ReLU)
Residual Block ×2	(3×3 kernels), 128 filters (BatchNorm + ReLU)
GlobalAvgPool Layer	1×1
Dense Layer	256 Neurons (ReLU + Dropout 0.25)
Output Layer	1 Neuron

C. Cross-Modal Attention with Auxiliary Depth Head

The final model built upon what was learned previously (see table III). The team looked for more advanced options which could improve performance, such as the vision transformer, which has gained a lot of attention in recent years [5]. While researching transformers, the team examined multi-modal models, which can ingest different data types (e.g. image and text). For this task, it means that the colour image and depth map can be treated as two separate "inputs" for an instance. Fundamentally, despite being in PNG format, the depth map data should not be treated the same as image data (i.e. extra colour channel). Doing so limited the extent to which the other models could perform well [6].

This cross-modal model processes the image and depth data in two separate streams, performing feature extractions using convolutional layers. The depth maps are harnessed to learn which locations in the images are important (effectively a form of attention). The model can then suppress or amplify based on a region's importance [7], it also directly fuses the combined

image and depth data - depth is no longer a separate channel [8].

This cross-modal fusion stage is repeated twice in different layers of the network to focus on spatial details at different scales [9], somewhat analogous to how YOLO ("You Only Look Once", a prominent real-time object detection system) has anchor points at various layers to allow for detection to occur at various scales [10]. The model ran for a maximum of 60 epochs, early stopping was enabled (patience of 8) and an initial learning rate of 3e-4 was used with the AdamW optimiser (improved version of Adam optimiser with better regularisation). This was the final model the team evaluated, as it performed strongly (see results section).

TABLE III
CROSS-MODAL ATTENTION MODEL WITH FUSION AND RESIDUAL BLOCKS

Layer	Layer Details
Input	RGB (3 ch) + Depth (1 ch)
Convolutional Layer Convolutional Layer	(RGB), 48 filters (BatchNorm + ReLU) (Depth), 24 filters (BatchNorm + ReLU)
First fusion stage	RGB: 48 filters + MaxPool 2×2 Depth: 24 filters + MaxPool 2×2 Attention: 1×1 Conv (Sigmoid) Fusion: Concat + Conv (3×3), 32 filters
Second Fusion stage	RGB: ResBlock + Conv (3×3), 64 filters Depth: ResBlock + Conv (3×3), 32 filters Attention: 1×1 Conv (Sigmoid) Fusion: Concat + Conv (3×3), 64 filters
Trunk Block GlobalAvgPool Layer	ResBlock (3×3, dil = 2) 1×1
Dense Layer Output Layer Auxiliary Head	256 Neurons (ReLU + Dropout 0.25) 1 Neuron (Regression) Conv (1×1) for depth reconstruction (H/4)

V. RESULTS

Results presented focus on the final model (Cross-Modal Attention with Auxiliary Depth Head). Including hyperparameter tuning etc. the final model had a total runtime of approximately 2 hours.

Due to limited compute, team chose to only test a single hyperparameter for the final model, the initial number of filters for the RGB channels (32, 48, 64). With 48 filters yielding the best results (32 filters worsened final MSE by ∼700, and 64 filters led to overfitting on validation data).

Regarding the final results, the team first investigated the performance of the cross-validation (i.e. had it managed to create representative splits of the data). Across the partitions, the team found that the mean Calorie values were between 229-234, indicating fairly uniform partitions and confirming that the final results were not unduly impacted by poor cross-validation.

To determine the degree of overfitting and how well the final model converges, the team examined the loss curve (see figure 3). In terms of comparing the training and validation losses, one notable trend was the greater instability and variability of the validation loss curves across the partitions - which can be ascribed to the comparatively smaller size of the validation set. The team was also pleased to see that the model converged and the low difference between the training and validation losses, indicating that the model is generalising well.

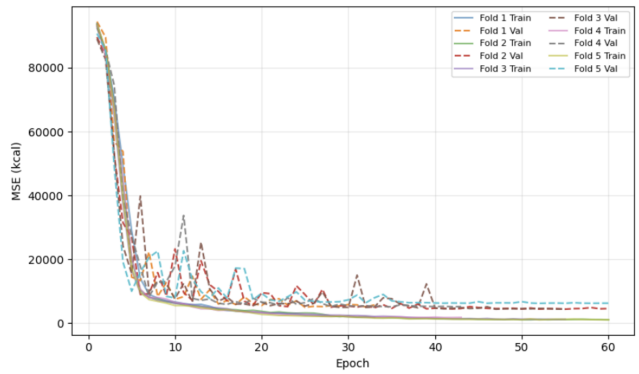

Fig. 3. Learning curve for final model

The final model had an overall validation MSE of 4993.62. To contextualise this result, the Mean Absolute Error (MAE), was 42.83, given that the average is heavily skewed by a few instances that were very poorly predicted (see figure 4), one can see that the majority of predictions were quite accurate - which is impressive given the complexity of the task.

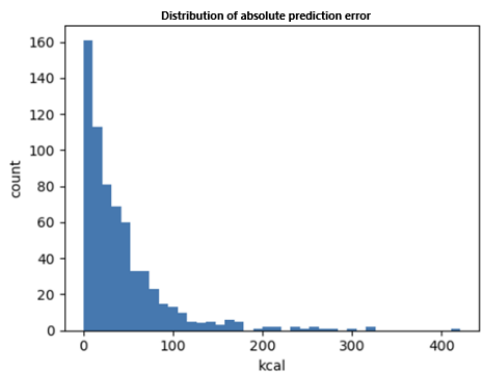

Fig. 4. Distribution of prediction error

For the test data (Kaggle competition), the model achieved an overall MSE of 4,758.33, further confirming that the final model used by the team is performing well.

VI. ERROR ANALYSIS

When examining the types of instances that the model was able to predict accurately, a clear trend emerges. Instances with only a single type of food (see figure 5) were able to be predicted quite accurately by the model, especially for low-calorie foods. The model generalised well when dealing with dishes under 500 Calories (majority of training data was within this range) in general.

Unfortunately, the model generalised quite poorly on complex dishes high in calories (see figure 6), leading to the team's decision to remove (high-calorie) outliers, this improved results, reducing MSE from 11,073.47 to 4993.62. There was also a mean bias error of 2.63 (indicating slight over-prediction on average).

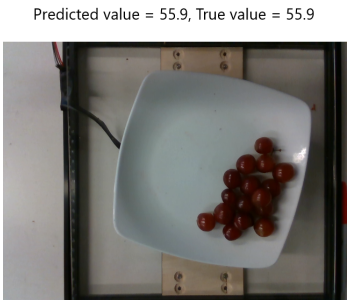

Fig. 5. Example of dish predicted accurately

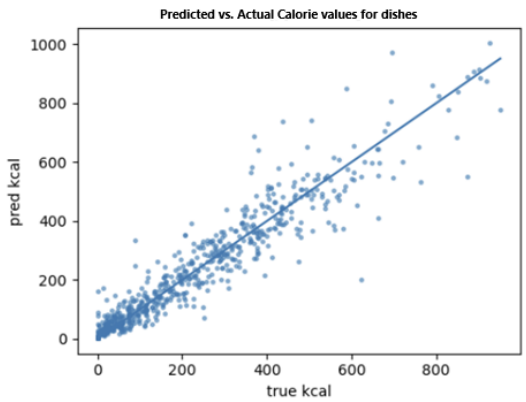

Fig. 6. Comparison of predicted and true value

The model's weakness with generalising to high-calorie dishes is both due to the complexity of classifying instances with multiple food types and the *relative* lack of training data for high-calorie instances (see figure 2). If one examines the Grad-Cam output (which highlights where the model was "looking"), one typically sees that in the high-calorie dishes the model is not looking in the proper locations - often focussing on the plate or background (see figure 7).

Fig. 7. Example of complex dish predicted inaccurately

VII. Limitations and Future Work

A. Limitations

A key limitation was the amount of compute available, constraining the complexity of the models that could be tested (e.g. could not test full ResNet architecture) and the breadth of testing (e.g. number of hyperparameters).

Each instance can contain multiple foods. Given that the task involves only predicting the overall caloric content of a dish, no per-item guidance is provided, which does curtail the usefulness of the predictions (e.g. some people may want to know what parts of the dish contribute the most to overall calorie content).

Another limitation is the fact that the training data images all appear to be from an overhead view - with most images being taken under optimal conditions. This would reduce the usefulness of the model were it to be used by people in real-world situations (various angles, image quality, lighting etc.).

B. Future Extensions

Arguably, with greater computing power, more complex model(s) could have been evaluated, potentially leading to better results. Additionally, a larger dataset could have been utilised (containing more high-calorie instances), improving generalisation for those dishes. Given that most standard meals are between 500-700 Calories at minimum (with fast food meals having even more calories) [11], improving the generalisation for high-calorie dishes is crucial if the model is to be useful in real-life settings.

Future models could also be developed which are aimed for use by people in real-word situations (e.g. smartphone app). This would necessitate additional training data (images from various angles, suboptimal quality etc.). Moreover, depth would have to be estimated (e.g. using MobileXNet [12], [13]) or a model would simply not use depth data.

Another interesting extension could involve predicting other nutritional information for dishes. Instead of just predicting calories, there could also be predictions for the amount of fat, protein, sugars, carbohydrates etc. All of this information is highly useful for nutritional purposes.

A more complex extension could be to add segmentation to a future iteration of the model, allowing a more granular (per-item) estimation - this could be very beneficial for predictions involving large dishes.

VIII. Conclusion

Estimating calories by using machine learning models using images has many positive applications (e.g. people trying to reduce calories through dieting). Through utilising the provided Nutrition5K dataset, the team iteratively refined their approach, beginning from a standard CNN architecture, adding residual blocks, before adding a cross-modal attention architecture with auxiliary depth input.

To ensure the results were robust, 5-fold stratified cross-validation and hyperparameter tuning were utilised. Through error analysis, it was found that much of the prediction error originated from multi-item dishes with a high caloric content, allowing the team to realise that removing extreme outliers would significantly improve performance.

The result (validation MSE 4993.62) indicates that the final model is performing well, demonstrating that a fairly modest architecture using cross-modal attention and auxiliary depth input run on a consumer grade GPU can act as a robust (and useful) caloric estimator.

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